

Evidence Inventory: Complete Overview of All Experiments

RQ		Judging	Target Variable	Comparison	Metric Type	Key Metrics	#	CLDs	Size	Breakdown	Runs	Prom. Mod.	Search	Quality	Path	
RQ1a: Synthetic Corruption	Synthetic Cor-ruption	Correct.	Corrupted vs Clean edges	Judge verdict vs corruption status	Correlation	r=-0.354 (p<0.01); Prec. 100%, Rec. 29-46%	2	Dep, Soc	155	Dep: 110, Soc: 45	1	1	1	—	INCOMP.	rq1.1
RQ1a: TruthQA		Correct.	Halluc. vs Truth.	Judge verdict vs GT	Classification	Acc. 89.7%, F1=0.900, Rec. 92.8%	N/A	TruthQA	1,634	H: 817, T: 817	1	1	1	—	STRONG	truth
RQ1a: Citation Providers		Cit.-based	Edge correctness	Judge scores across providers	Aggr. score	PubMed 0.306, Brave 0.296, Perp. 0.278, OpenAlex 0.232	1	Ulemans	600	150 × 4 prov.	1	1	1	4	GOOD	judg
RQ1a: Human Valid.	Human Valid.	Cit.-based	Judge-Human agree.	LLM vs expert	Inter-rater	κ=0.618 (Sub.); Agr.=80.6%; r=0.732	2	Dep,Soc	72	30+30+12; 2 CLDs, 2 gens	3	1	2	—	STRONG	comb
addlinespace RQ1a: TP vs FP		Correct.	TP vs FP edges	Judge score TP vs FP	Mann-Whitney U	FP=0.873 vs TP=0.869, p=0.52 (n.s.), d=0.015	3	Dep,Soc,Old	1,394	TP=42, FP=161, TN=1126, FN=65	1	1	1	—	STRONG	rq1.1
RQ1b: Corrector	Corrector	Correct.	Edge validity vs GT	Before vs After correction	F1 improve.	ΔF1 +0.012 avg (Soc +0.030, Dep -0.006)	2	Dep, Soc	155	Dep: 110, Soc: 45	1	1	1	—	WEAK	rq1.1
RQ2: Metrics	CI	N/A	TP vs FP edges	CI metrics as predictors	AUC (discrim.)	Best AUC=0.564; TP=88, FP=319, FN=126, TN=2252	3	Dep,Soc,Old	2,785	D:364, S:180, O:2,241	2	1	1	—	STRONG	exp.1
RQ2: Ensemble	Ensemble	N/A	TP vs FP edges	Ensemble classifier	AUC (discrim.)	AUC=0.725 (Logistic); Gen-only 0.695	3	Dep,Soc,Old	1,373	From RQ1 base	1	1	Mult.	—	GOOD	rq2.1
RQ2: Nested CV	Nested CV	N/A	TP vs FP edges	Nested CV valid.	Unbiased AUC+CI	AUC=0.624 [0.478, 0.824]	3	Dep,Soc,Old	219	Subset valid.	5×3	1	1	—	STRONG	rq2.1
RQ2: Cross-Domain	Cross-Domain	N/A	TP vs FP edges	Train 2, test 3rd	Leave-one-out AUC	AUC=0.51 (random)	3	Dep,Soc,Old	210	TP+FP runs) (9	Var.	1	1	—	STRONG (neg)	exp.1
RQ3: Deep Research	Deep Research	Cit.-based	Evidence detection	TP vs FP vs FN rates	Detect. %	TP 70.5%, FP 62.4%, FN 42.9%; gap 8.1pp	3	Dep,Soc,Old	285	O:184, D:84, S:17	1	1	1	Mult.	MODER.	Mult

Legend: Green = Strong/Good evidence; Orange = Incomplete/Moderate; Red = Weak/Undocumented.

Abbrev.: GT = Ground Truth, Correct. = Correctness, Cit. = Citation, Dep = Depressive, Soc = Social Norms, Old = Older Persons, Aggr. = Aggregate, Prom. = Prompts, Mod. = Models, Mult. = Multiple, Prec. = Precision, Rec. = Recall, Acc. = Accuracy, Perp. = Perplexity, H = Hallucinated, T = Truthful, Var. = Varies, n.s. = non-significant.